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EDITORIALS

Making Sense of the Digital Automation of Education

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Introduction

The ongoing digitization of education is ushering in all manner of changes to schools, colleges, and universities—not least the absorption of small (often imperceptible) automations into everyday educational practises and processes. Now, for example, the default setting of a learning management system might be to automatically cross-check any submitted written assignments with plagiarism detection software in order to generate a measure of 'originality'. Some learning management systems might then link to automated-essay-grading software to quickly produce a set of provisional scores for a human grader to later orientate their own judgements against. At the same time, students writing these assignments might have incorporated neat turns-of-phrase suggested to them by their word processor 'autocompletion' feature, and might even have also run their writing through another plagiarism detection system to gauge the chances of being caught. Elsewhere, many classrooms have 'student activity monitoring' software running in the background—monitoring keystrokes and online activity to judge if classes are actually engaged in their work or not. Whereas all these small tasks of checking, noticing, choosing, and deciding might once have been the responsibility of a school teacher or university tutor, now they are increasingly delegated to technology.

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Such automations have become quickly woven into the digital fabric of schools and universities to the point that they now pass unnoticed. This Special Issue of *Postdigital Science and Education* challenges us to pause and pay renewed attention to these automations. The papers in this issue prompt us to ask what is going on when such tasks, while they may seem small, are delegated to software. These papers also prompt us to question the reasons why digital automations of education are being enacted, with what gains, and with what losses. While such questions might seem obvious, this has to date proven a curiously uncontroversial and non-contested area of education. On one hand, this lack of controversy reflects the seemingly small scope and limited significance of many of these automations, as well as the incremental nature with which they build up. Yet, on the other hand, there are also understandable reasons for educators to be broadly enthusiastic about this technology. For example, many educators have so far been happy to welcome technological advances purely on the grounds of presumed efficiencies and timesaving. Conceived along these lines, there is perhaps much to be hopeful about—not least the promise of digital technologies taking on some of the repetitive, mundane busy-work that many educators feel have come to blight their working lives. Why not let algorithms 'free up' teachers to concentrate on the higher-level pedagogical work that they have been trained for?

Running alongside these labour-saving arguments is the suspicion that automated technologies might actually be capable of doing a better job that a human teacher. For example, automated essay grading software can certainly process student assignments at an unimaginable scale and speed in comparison to even the most adept human grader. There is also understandable hope that many processes might be executed more fairly and more consistently than is possible when left to human educators alone. As Marion Fourcade (2022) puts it, this is the hope that 'the divination of the algorithm [can] replac[e] the arbitrariness of the bureaucrat'. While many readers of a journal such as *Postdigital Science and Education* might feel that such arguments are highly contestable, we should not discount them altogether. In many cases, digital automations will faithfully re-instantiate the steps and rules already instantiated in the forms, frameworks, and guidelines of the existing bureaucracy. As with all technological innovation, digital automations of education are not a wholly bad thing and, in many people's eyes, are certainly no worse than the non-automated arrangements that precede them.

The broad acceptance of these technologies to date reflects a general expectation amongst most education professionals that the full automation of pedagogical work (and therefore the complete replacement of teachers by machines) is extremely unlikely to occur. Most educators remain confident that their work is too complex and contextual to be completely taken on by automated digital tools. Certainly, while some routinised factory workers might well have cause to fear what Aaron Benanav (2020) describes as full 'lights out' automation (i.e., workplaces requiring no human presence so that the lights can be turned off), most people would agree that schools, colleges, and universities continue to be sites of intricate knowledge work that is centred around relational, embodied human activity. As such, it makes sense to anticipate the partial automation of schools and universities—populated by what Benanav describes as 'labour-augmenting' technologies. Most people in education, therefore, remain fairly comfortable that extreme predictions of fully automated

classrooms and 'robot teachers' remain way off the mark. Instead, it is generally accepted that the challenge for contemporary education professionals is figuring out how best to work alongside (or ideally, work along with) these technologies.

That said, we need to avoid making the mistake of framing these discussions purely in terms of teachers and teaching. To date, the most pervasive forms of educational automation relate to the administration, management, and governance of education. For example, recent years have seen the Indian Ministry of Education implement computer-based automation to administer the country's vast annual parent-school application procedure and filter out erroneous cases. Similarly, university admissions teams around the world now routinely make use of algorithmic sorting to draw up initial long lists of prospective student applications. Of course, many of the decisions now being delegated to digital systems are far more mundane than these potentially life-changing examples. For example, newly built school campuses are now often fitted with heating and lights that switch on under certain conditions, and visitor management systems that recognise the faces of enrolled staff and students while denying entry to strangers. In this sense, any talk about the 'automation of education' covers a wide range of automations that have been developed for a wide range of purposes and are resulting in a wide range of consequences. Thus, our conception of automation needs to stretch well beyond the particular domain of teaching and pedagogy, and also take note of arguments arising around other institutional, organisational, and social contexts outside of the school and university.

Questioning the Efficiencies of Education Automation

In one sense, the digital automations covered in this Special Issue might simply be rationalised as a necessary aspect of education systems keeping up to date with the rest of society. After all, similar automations are increasingly evident across many other areas of everyday life and society. Yet, the papers in the Special Issue convey a collective warning against taking these promises of automation in education at face value. It is well worth considering how and why the realities of these automated technologies might not always be fully meeting expectations. Such scepticism should not be too difficult a step for most people to take. Indeed, the fallibilities and frailties of digital automation should be obvious to any smartphone owner who has experienced their device's autocorrection and autocomplete features suggesting wildly inappropriate or insensitive phrasing. More seriously still, we have become accustomed to reports of facial recognition systems having a hard time recognising people of colour. Such glitches, errors, and failures have been long noted in education. For example, Les Perelman's work has shown how carefully arranged screeds of gobbledygook can nevertheless be awarded top marks by automated-essay-grading systems (see Perelman 2022). We ignore such shortcomings and omissions at our peril. These are not simply annoying episodes or amusing anecdotes of technological teething troubles.

Of course, from a purely technical point of view, it could be countered that these failures are all acceptable (if not expected) features of the advanced forms of machine learning that lie behind contemporary digital automation. After all, human judgement and decision-making are not perfect either and, in simple terms, machine learning involves the development of mathematical models that

are refined when confronted with errors. In effect, these systems are designed to fail on a large scale in order to improve. This kind of machine learning has prevailed as a form of AI over more conservative or 'tame' forms built around knowledge representations such as intelligent agents where human expertise and wisdom is systematically modelled. As such, many of the automated technologies currently being installed into schools, colleges, and universities are certainly not elegant, sensitive intelligent representations of human expertise and action. As Hillary Mason (2018) reminds us, these technologies do not somehow constitute a form of 'inscrutable magic'—rather this is technology that is the product of 'math and data and computer programming, made by regular humans'.

However, in the case of deep learning, which can be argued to be the prevailing form of machine learning of the 2020s, there is an inscrutability to the internal processes of the mathematical models generated that can give a misleading impression of 'more-than-human' sentience. Instead, it is important to see these seemingly inscrutable processes as the aggregation of sometimes hundreds of relatively simple and understandable operations. The output of each of these operations is trained through 'brute force' trial-and-error working with massive datasets and together they produce what appears to be a sophisticated result that is evaluated in relation to the assumptions made by developers. In this sense, the inscrutability of the technology derives from the sheer number of simple operations, and not from any sort of magical insight.

This spirit of statistically-relentless and socially-insensitive computation is exacerbated by the commercially-driven nature of how these technologies are sold and rolled-out into educational settings. In his interview piece for this Special Issue, Mark Andrejevic (in Andrejevic and Selwyn 2022) calls out education's 'almost incomprehensible readiness' to hand control over to the private actors who design and market automated decision-making software and systems. Of significance here, then, are the ways that subsuming authority to automated technologies opens education up to a number of prevailing 'Big Tech' logics. For example, in a commercial sense, any person coming under the gaze of an automated system (regardless of that person's background or circumstances) tends to be framed as a 'user' operating wholly within a bounded digital ecosystem. This also implies a 'politics of scaling'—in other words, the ambition to develop education automations that move beyond localised actions, and instead develop into scalable, universalised actions that can dominate markets on a society-wide basis (Pfotenhauer et al. 2022).

Also of significance is what can be described as the Big Tech 'politics of modularity' (Birch and Bronson 2022)—where technical standards for interoperability are exploited so that any external automated service can be easily 'plugged in' to a school or university's broader digital infrastructure, creating a logic of just-in-time, unbundled automation where specialised services and resources are assetized and rented out on an on-demand basis (see Hansen and Komljenovic 2022). Indeed, one argument that emerges across this Special Issue is that such commercially-driven logics fundamentally clash with what might be traditionally seen as public education values and logics—promising a new form of education based around principles of contingency, compatibility, standardisation, and consumer-led provision of 'education as a service'. In this sense,

digital automation might not simply be a benign, neutral addition to existing ways of doing education. Instead, much digital automation presumes a radically reconfigured form and function.

Holding the Automation of Education to Account

Such is the scale and scope of the ongoing digital automations of education that our discussions urgently need to move beyond simply speculating on whether all of this is a 'good thing' or not. Given the extent to which these technologies are already being used, it is perhaps best to start reflecting on how we might want to best live with, gain control of, and/or fight against these technologies. As such, the most pressing debates to be having at this juncture are pragmatic discussions around technological regulation, governance, and even resistance. This reflects what Frank Pasquale (2020) has termed the 'second wave' of algorithmic accountability, which involves raising questions around the responsible integration of these systems in society. Indeed, Pasquale's 'new laws of robotics' offer an excellent starting point for such discussions—a challenge first outlined in Pasquale's interview for this Special Issue (Pasquale and Selwyn 2022) and subsequently picked up in the group discussion paper coordinated by Mathias Decuypere and colleagues (Decuypere et al. 2022).

Indeed, many of the papers in this Special Issue stress the need to begin asking difficult questions of these technologies—refining expectations of automated education as a problematic to be investigated, rather than a problem to be solved. In other words, discussions of automation in education need to be deliberately pushed toward fundamental questions after Emily Bender (2022) of what is being done in education in the name of 'automation', to whom, who benefits, and how democratic oversight might be exerted. In this spirit, then, we can conclude by considering a few different lines of thought that emerge from the papers in this Special Issue, and along which we might want to continue to hold automated technologies to account.

The Harms of Educational Automation

First and foremost in holding automated technologies to account is the need to highlight instances when automated systems result in obvious social harms within educational contexts. As is now widely acknowledged, some of the most substantial harms arising from digital technologies over the past few years have occurred when AI models amplify the discriminations baked into their training data and subsequently drive autonomous systems to discriminatory and disadvantaging ends. Various instances of this have already occurred in education, such as voice analysis systems erroneously judging students with non-native accents to be cheating on English-language tests, and exam prediction models consistently computing higher automated grades for students attending fee-paying schools (e.g., National Audit Office 2019).

Other harms might appear less substantial, yet nevertheless also need to be taken seriously. Take, for example, reports of online exam proctoring systems failing to detect the faces of Black students, or non-binary students having to mis-identify

themselves as either 'M' or 'F' in order to register a 'valid' system profile. While such exclusionary glitches might not be designed deliberately into technologies, they effortlessly replicate and reinforce discriminations that minoritized students encounter regularly throughout their engagement with education systems. All told, perhaps the most pressing matters to be discussing about the roll-out of autonomous technology throughout education relate to how any data-driven system is imbued with 'a tendency to punch down: that is, the collateral damage that comes from its statistical fragility ends up hurting the less privileged' (McQuillan 2022: 35).

Crucially, it is also important not to presume that these harms can somehow be 'fixed' and made 'fairer' through the curation of more inclusive training datasets, or more mindful approaches to progressive software design. Instead, educational discussions around this technology need to take heed of growing arguments—led by scholars such as Joy Buolamwini, Safia Noble, Simone Browne, Ruha Benjamin, and Timnit Gebru—that frame digital automation as exacerbating forms of 'engineered inequality' (Benjamin 2019) in already inequitable social contexts. This points to the ways in which AI technologies result in inevitably oppressive and disadvantaging outcomes 'given their design in a society structured by interlocking forms of domination' (Benjamin 2019: 47). Following this logic, it is highly unlikely that automated systems implemented in already an unequal or discriminatory education context will somehow lead to radically different liberatory and emancipatory outcomes. Instead, it is most likely that any automations will lead to amplifications and intensifications of existing tendencies and outcomes.

This also raises the point—evident in a few of the papers in this Special Issue (e.g., Swist and Gulson 2022; Gibson 2022)—that harms associated with automation in education are profoundly relational in nature, and therefore likely to be experienced differently according to individuals' different backgrounds and circumstances. Any instance of some people being disempowered and disadvantaged by the implementation of digital automation in education is accompanied by other people being empowered and advantaged. As such, any particular automated technology might appear to work perfectly well for many teachers and students. Nevertheless, for many others, the same technology can simultaneously be experienced in profoundly less advantaging ways. This raises the need for discussions to progress beyond broad-brush concerns over presumed forms of universal harm arising from AI technologies—for example, what Viljoen (2021) terms 'dignitarian' fears over a general 'dehumanisation' of education processes and practises. Instead, more detailed questions need to be asked about the localised harms being experienced by specific individuals and groups. Regardless of our own personal positive experiences, we need to remain aware of the embedded socially differentiated ways in which education automations can impact on others in harmful and deleterious ways.

The Reductiveness of Educational Automation

Second is the need to pay close attention to the aspects of education that are diminished, de-emphasised, and side-lined in any educational turn to automation. More specifically, this requires us to problematise claims that it is

possible to statistically model, monitor, and compute educational events without the inherent reduction of key qualities and characteristics. Of course, most technology developers would contend that it is possible to construct 'epistemologically sound' mathematical representations of any aspect of education as it is experienced and engaged in by students and teachers. For example, automating a bounded situation in a school or university 'smart campus' might require a small set of logical steps—an authorised student approaches a door, they are recognised by a smart-camera, and the door opens. However, it can be reasonably contended that there are many educational events and situations that are not possible to codify and model in this neatly defined and bounded manner. For example, while some aspects of the judgements that go into evaluating a piece of written work might be readily automated, the breadth of human judgements and expertise that lie behind the act of grading a written assignment might be reasonably seen as irreducible to even to the most contextually rich, complex statistical model.

At best, then, many automated systems in education are inevitably programmed to make narrow decisions on the basis of limited 'proxy' variables and, therefore cannot be seen to be fully sensitive to the social context of the process or practise being automated. As such, it might be argued that the data used to drive digital automations can never constitute an epistemologically sound representation of any human condition but are instead simple 'signifiers whose referent is no longer a human subject but a cluster of correlations' (Bolin and Andersson 2015: 4). The concern here is that any instance of arranging and enacting a 'real life' education situation through computer-based models and simulations of the real-life situation will inevitably alter the fundamental human experience of education. As a result, one's accumulated encounters with these various constrained automations begin to comprise a notably 'narrow bandwidth' experience of schooling.

For some, this reductiveness is a fatal flaw in the current turn to digital automation. In short, it can be argued that these are systems that promise short-term effectiveness in narrow application domains while being defective at a more fundamental level of human sociality. Of course, some people remain happy to anticipate AI and other automated technology fast advancing to a state of being somehow capable of a 'framelessness' (Andrejevic 2020)—the fantasy of a perfectly complete, totally comprehensive '1:1' capture of all human experience that then can be used to drive fully sentient technologies. Even if this framelessness were possible, it side-steps some basic aspects of human thought—such as Karl Popper's (1994/2001) observation of the inherent provisional and unreliable nature of human experiences, along with the fact that knowledge arises from the capacity to make inferences from information rather than simply having the information to hand. As such, the fantasy of 1:1 data does not do away with the human need to make inferences (that is, to impose 'frames'), but simply replaces human forms of inferencing with computer-driven forms of inference. These are not more effective, efficient, or free from bias, but more often opaque and just as likely to be coloured by the biases involved in human inferencing. As implied in recent critiques of automated language systems as 'stochastic parrots' (Bender

et al. 2021), we automate complex human processes such as sense-making and meaning-making at our own risk.

The Losses Inherent in Educational Automation

Another important question that needs to be asked of digital automation in education, is one of loss—as Moser et al. (2022) put it, 'what do humans lose when we let AI decide?'. In his interview piece for this Special Issue, Mark Andrejevic (in Andrejevic and Selwyn 2022) frames this in terms of Harold Innis' (1951/2008) notion of 'media bias'—i.e., how any automated technology that is chosen to be used in a classroom will be designed around particular time/space arrangements, forms of knowledge, and sets of power relations. In this spirit, it is worth considering a number of possible reductions, elisions, and substitutions when it comes to the automation of education.

One key loss is what might be described as the replacement of judgement with reckoning. As Moser et al. (2022) reason, dominant justifications for the deployment of automated systems in social settings tend to conflate reckoning (i.e., decision-making based on summing up of various kinds of data, computation, and rule-driven rationality) with judgement. Genuine human judgement, Moser argues, is dependent on reasoning, imagination, reflection, and empathy. This is echoed in Andrejevic's contribution to this Special Issue that the non-predictability of human judgement sets it apart from machine-based automated decision-making (Andrejevic and Selwyn 2022).

Such concerns are illustrated, for example, in the ways in which automated essay grading software essentialises what a human essay grader does in a mechanised procedural manner. As Siddarth et al. (2021) put it, this is technology that 'mechanises our conception of human capacities'. Yet, as Wagener-Böck et al. (2022) argue in this Special Issue, when examined in practise, much of what is conceptualised as automation in education is actually a form of 'symmation' where humans and technology co-produce the effect of automation. In this sense, the technology not only conceptualises human capacities in mechanistic ways but also shapes practises so that the conception is performed. As is suggested throughout this Special Issue, even the smallest tasks carried out by teachers and students in educational settings have important social and relational qualities that are dispensed with when these actions are codified as 'tasks' and automated accordingly.

A similar conflation might be observed in terms of the replacement of discovery with retrieval —the former depending on an element of serendipity and fortuity that is not present in the 'recommender systems' that point us to already known content, driven by correlations and patterning. As Dwayne Monroe (2021) contends, applying statistical methods to already collected data is of no use in terms of developing capabilities such as inquisition, intuition, and taste-making. At best these systems are 'mirroring your past behaviour back at you, making it harder to find something truly new'. As several of the papers in the Special Issue also suggest, we, therefore, need to question the assumptions taken in the development of automated systems in relation to their outputs and effects on education. To do this, we need to resist the temptation to get caught up in the apparent precision of a system's data visualisations, the certainty of its reporting, and/or the definitiveness of

the predictions that it purports to provide teachers and students. Indeed—to paraphrase Knott (2022)—'the algorithms driving these apps give the impression that they see the *essence* of [education] and render it accurately. But we're only really seeing it through the financialised language of their creators'. (emphasis from the original)

Seeing Educational Automation in Terms of Power

Following on from these latter points is the need to explicitly frame the automation of education as entwined with dynamics of power and control. This pushes us to move beyond socially concerned talk of 'harms' and similar notions of (dis) empowerment, to deliberate conversations around the politics of automation that are imbued with a sustained deliberation of power. Here, we might follow Verdegem's (2021) focus on the symbolic, political, and economic forms of power implicit in AI and digital automation. Take for example the symbolic power that derives from meaning-making and influencing the actions of others. In terms of digital automation, this is certainly evident in what Verdegem terms 'AI ideology'—i.e., the ways in which human consciousness is manipulated to see digital automation as an important (if not inevitable) means of determining future forms of society and/or economy. Current dominant forms of AI ideology range from vague progressive notions of how digital automation might foster new forms of a 'good society', through to the boosting of fully automated work as the basis for a new era of 'digital capitalism'. How particular voices and interests are able to drive these different societal understandings of what education automation is (and what education automation is for) is therefore a key form of contemporary power that we need to pay close attention to.

At the same time, digital automation is clearly linked with political power that derives from having the authority to coordinate individuals and their interactions. This political power is evident in what Verdegem (2021) terms the 'social practises'of AI. These include the ways in which developers, programmers, and engineers choose to select training data and design the classification systems, models, and procedures that drive processes of AI and more specifically machine learning. Who gets to be involved in these design processes, who gets to determine what is included/excluded in the development of these systems (e.g., how things are represented through data), and what value is then extracted from their continuous refinement, are therefore all key forms of political power that are fast coming to bear on education.

Clearly, these technologies are also entwined with forms of economic power—power that derives from accumulating resources for productive activity. In terms of the digital automation of education, this form of power is perhaps most obviously exercised through ownership of the data infrastructures and computational power that underpins the development and deployment of AI systems and automated-decision-making technologies in education contexts. This form of power also lies in the expert data science and human resources involved in the development of these technologies. As such, a number of papers in this Special Issue (e.g., Gilliard and Selwyn 2022; Andrejevic and Selwyn 2022; Hansen and Komljenovic 2022) make the point how the digital automation of education involves the centralization of power and the commercial agendas of

IT industry and 'Big Tech' actors, alongside a corresponding sidelining of education institutions, educators and other stakeholders. This chimes with Birch and Bronson's (2022) argument that we need to give careful thought to how firms such as Amazon, Microsoft, and Google act as powerful 'gatekeepers'—both in economic terms and in terms of shaping the digitization of public services such as education. Therefore, there is much to discuss here—from how the educational ambitions of Big Tech actors might be regulated and curtailed, through the need to politicise matters of digital infrastructure and resource ownership (Jäger 2019).

A number of papers in this Special Issue also explore the specific contention of how the outcomes of automated technology fold into state and structural power (e.g., Swist and Gulson 2022; Gilliard and Selwyn 2022). This chimes with the description emerging from various critical commentators of digital automation as a key part of the 'synthetic governance' of education—a form of control that does not pursue human replacement, but standardisation, docility, and the flattening of practise (see Gulson et al. 2022). This is evident, for example, in the ways that automated systems demand a legibility from teachers, students, and classrooms—i.e., outputs and outcomes that are 'machine readable' and 'parseable'. As Legacy Russell (2021) reminds us, 'demanding legibility is a strategy of the state'. The legibility demanded of teachers and students from education technologies, therefore, infers a mode of continuous accountability and possible coercion to act differently.

As Andrejevic notes in his interview piece (Andrejevic and Selwyn 2022), the logic of behavioural economics and 'nudging' inherent in automated media suppose that people's behaviours can be internalised in ways that conform to a bounded model of the algorithmically rendered classroom. In this sense, automated technologies are concerned with exercising power over the actions and activities of the teacher, rather than the complete replacement of the teacher. As Anton Jäger (2019) concludes, any talk of the total replacement of the teacher, therefore, constitutes an act of class discipline—the threat of 'we can replace you at any moment!' needs to be seen as a disciplinary move rather than an actionable possibility.

Acknowledging the Futures and Histories of Educational Automation

A final set of ideas worth extracting from the Special Issue contributions is the call to engage both with the possible futures of digitally automated education and the preceding histories of pre-digital automation. To date, academic commentators have perhaps proven most comfortable with speculating and engaging with the futures of digital automation. Indeed, various papers in this Special Issue raise a number of pertinent points around continuing these future-facing discussions. For example, as papers by Cerratto Pargman et al. (2022) and Robinson (2022) show, there is merit in engaging with the many various predictions, speculations, and expectations around the possible future automations of education—not least in highlighting how talk of the imminent automation of education acts as a means of providing commercial actors and commercial logics with legitimacy. Indeed, from a sociology of expectations perspective, engaging with corporate and policy predictions can be a generative way to examine how corporate actors use ideas of the

future to 'guide activities, provide structure and legitimation, attract interest and foster investment. They give definition to roles, clarify duties.' (Borup et al. 2006: 285–286) As such, some of the papers in the Special Issue give a strong sense of the 'sociotechnical imaginaries' around education automation being advanced by governments, corporations, and other stakeholders in the ongoing digitization of education.

Nevertheless, we need to resist any temptation to play along too closely with the dominant forecasts and predictions that inform mainstream understandings and conversations about possible digital futures. Recently, Lee Vinsel (2021) has derided the 'academic business model' of critical researchers 'playing along with hype to score cash money and prestige'—what Vinsel curtly terms 'lending credibility to industry bullshit'. This chimes with Alfred Nordmann's (2007) concerns with social scientists who develop critiques of 'if/then' scenarios that engage credulously with dramatic corporate claims over 'incredible futures' driven by 'technological hubris'. The dangers of engaging with this type of speculative talk around education automation are obvious. Engaging critically with corporate expectations of automated futures runs the risk of reproducing (and even increasing) hyperbole, and lending credibility to industry promotional claims merely by taking them seriously, overstating the abilities of tech firms and/or the capabilities of their emerging products. As well as compounding unrealistic expectations, Nordmann (2007) warns that work of this nature distracts attention away from already-existing concerns—especially the actual risks and harms highlighted previously in this paper arising from current automated technologies.

Contributing to disrupting hyperbole around the potential benefits and risks of automation, other contributors to the Special Issue, therefore, highlight the need to pay more attention to the many historical precedents to the contemporary automations of education. All the ongoing talk of automatic exam grading, personalised learning, and bureaucratic efficiency has a history that needs to be acknowledged and learnt from. Indeed, prevailing claims, hopes, and fears around the general automation of society in the 2020s are not far removed from the claims, hopes, and fears in the 1950s—such as robots taking jobs, fears over dehumanisation, and hopes for increased precision and safety in the workplace.

Looking at the histories of automation of education, therefore, adds an invaluable dimension to our understandings of the area. As Bergviken Rensfeldt and Rahm (2022) demonstrate, paying close attention to the politics and technologies of education automations over time, renders visible societal debates, concerns, and resistances that were once prominent but that have since diminished. Revisiting and revitalising these historical discourses can, therefore, shine new light on education automation as it is currently materialised. Looking back over twentieth-century examples from the early attempts to 'computerise' classrooms in the 1970s and 1980s, there are many examples of questionable or malfunctioning automations that offer valuable insights on issues such as social justice and fairness (c.f. Park and Humphry 2019), as well as on issues around the values of public education and the roles of pedagogical work.

Conclusions

The contributions across this Special Issue combine to give a strong sense of the continued need to tackle the topic of automation and education in socially nuanced, politically aware, and appropriately sceptical ways. These forms of technology—and the logics that underpin them—are now a significant aspect of the digitization of education, and therefore require our sustained attention. As with most aspects of the algorithmic landscape of everyday life, these are technologies that are quickly becoming woven into the digital infrastructure of education—and therefore increasingly 'invisible and forgettable' (Knott 2022), only revealing themselves when they impact on our lives in notably unfair, harmful, and obtrusive ways. As such, now is the time to be drawing full attention to the reconfigurations of power that are taking place in the name of automation. These are technologies that need to be the subject of ongoing conversations about what we want AI and automated decision-making in education to be, what purposes we can benefit from and are prepared to devolve to technology, and what aspects of education we are definitively *not* willing to hand over.

All the papers in this Special Issue, therefore, highlight the need to move these conversations away from bald, technical issues of precision, efficiency, speed, and scale. Instead, we need to better acknowledge how the claims and counterclaims around digital automation in education index deeper impulses and desires—around the forms of power, control, and standardisation that people wish to see established in education. Yet these conversations cannot be too dismissive or overly general—slipping from a constructive scepticism to a destructive cynicism of any technological application regardless of the details. As with all critical takes, there is little to be gained from getting bogged down in a technological pessimism amidst the prospect of education automation—seeing all things digital as inevitably broken, and expressing no desire to transform this predicament.

This, therefore, calls on researchers to take an active stance by developing better understandings of how digital automations can be meaningfully integrated into education and adopting a sense of realism (rather than idealism) when reflecting on where we might like to be going next (see Gallagher and Breines 2022; Gibson 2022). These are technologies that cannot be dis-invented or effectively banned outright. Given that digital automation is already a reality in a multitude of deceptively mundane guises, perhaps the most pressing questions to be asked are disarmingly practical—what do we want automation in education to be, and what is required to make those automations a reality?

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